

Laboratory investigations for the diagnosis of BP

Confirm the diagnosis of BP. The diagnosis is based on a combination of criteria encompassing clinical features, compatible light microscopy findings, and positive direct immunofluorescence microscopy (DIF) findings. The following steps are recommended for the diagnosis and classification of BP:

- Detection of circulating IgG anti-basement membrane zone (BMZ) autoantibodies by indirect immunofluorescence (IIF) microscopy using NaCl-separated normal human skin;
- Detection of anti-BP180 NC16A IgG autoantibodies and/or anti-BP230 IgG autoantibodies by ELISA;
- For detection of circulating IgG anti-BMZ autoantibodies, novel 'multivariant' IIF-based assays using multiple antigenic substrates are also recommended. In these BIOCHIP mosaic assays, various antigenic substrates are combined.

In rare cases of BP where circulating anti-BMZ antibodies are not detectable by IIF or commercially available ELISA, additional tests should be used to increase diagnostic sensitivity and to exclude other autoimmune diseases affecting the dermoepidermal junction, particularly anti-p200 pemphigoid or epidermolysis bullosa acquisita (EBA).

Histopathology

Specimens for light microscopy should be taken from early bullae arising on erythematous skin and placed in formalin solution. Typical findings include subepidermal bullae containing eosinophils and/or neutrophils, associated with a dermal infiltrate of eosinophils and/or neutrophils, or a margination of eosinophils along the dermal-epidermal junction. However, in non-bullous forms

or in the absence of blistering, histopathological findings may be nonspecific, such as eosinophilic spongiosis.

Direct immunofluorescence microscopy

Direct immunofluorescence (DIF) studies are the most critical diagnostic test—positivity is essential for confirming BP. It is recommended to obtain the biopsy specimen for DIF from perilesional skin, defined as erythematous nonbullous skin or normal-appearing skin within 1–2 cm of a lesion.

For transportation, skin biopsy specimens should be placed in either 0.9% NaCl solution, a cryotube with liquid nitrogen, or Michel's fixative. Alternatively, for storage and transport, 0.9% NaCl (with processing required within 24–72 hours), liquid nitrogen in a cryotube, or Michel's medium may be used.

- DIF typically shows linear deposits of IgG and/or C3 along the dermo-epidermal junction; occasionally, IgA and IgE may also be present in a similar pattern;
- Analysis of the n-serration pattern in DIF can be helpful and specific when combined with IIF to differentiate BP from EBA: an n-serrated pattern is suggestive of BP, whereas a u-serrated pattern is typically seen in EBA.